1. Business Problem

A retail/liquor distribution business buys from many vendors and sells across multiple stores. Leadership doesn't have a clear, data-backed view of:

- Which vendors are actually **profitable** to work with once real purchase cost (not retail price), freight, and slow-moving stock are accounted for
- Which vendors cause **inventory inefficiency** (stock sits too long, ties up capital)
- Whether **freight cost** is eating into margin on specific vendors
- Where the business is **overpaying or under-negotiating** on purchase price vs. what it sells for

Right now, purchasing decisions (who to order more from, who to renegotiate with, who to drop) are made on gut feel / vendor relationships, not on numbers.

2. Objective

Build a Vendor Performance Analytics solution that lets stakeholders (Purchasing Manager, Finance, Ops) answer, for any vendor or brand:

- What's our actual gross profit and margin from this vendor?
- How fast does their stock turn over — are we sitting on dead inventory?
- What's freight costing us as a % of what we buy from them?
- Who are our top 10 vendors by profit, and who are we losing money on?

Deliverable: a governed SQL data layer → a reproducible Python analysis (cleaning, KPI engineering, statistical validation) → an interactive Power BI dashboard with DAX measures, plus a written recommendation.

3. Stakeholders

Role	What they need
Purchasing Manager	Vendor-by-vendor profit & freight cost ranking, to guide renegotiation
Finance	Accurate margin numbers (based on actual purchase cost, not list price)

Inventory/Ops	Stock turnover & dead-stock flags, to reduce holding cost
Leadership	Executive summary: top/bottom vendors, $ impact of recommendations

4. Data Sources

File	Grain	Feeds
begin_inventory.csv	Store-Brand, Jan 1 2024	Opening stock & value
end_inventory.csv	Store-Brand, Dec 31 2024	Closing stock & value → turnover calc
purchase_prices.csv	Brand	Retail price, purchase cost, vendor, classification
purchases.csv	PO line item (2.37M rows)	What was ordered/received, actual cost, dates
sales.csv	Sale transaction (12.8M rows)	What was actually sold, revenue, excise tax
vendor_invoice.csv	Invoice (5.5K rows)	Freight cost, approval, PO/pay dates by vendor

5. Tools & Tech Stack

Layer	Tool	Purpose
Data storage & prep	Microsoft SQL Server	Single source of truth; heavy joins/aggregation at scale done here, not in Python
Analysis & validation	Python (Jupyter Notebook, VS Code, .ipynb)	Pull SQL Server results, engineer KPIs, run stats tests, EDA plots
Visualization	Power BI + DAX	Interactive dashboard for stakeholders

6. Key Metrics (KPIs) We'll Build

- **Gross Profit** = Sales Dollars – Purchase Dollars (actual cost)
- **Profit Margin %** = Gross Profit / Sales Dollars
- **Stock Turnover** = Sales Quantity / Average Inventory ((Begin + End)/2)
- **Sales-to-Purchase Ratio** = Sales Dollars / Purchase Dollars
- **Freight Cost %** = Freight / Purchase Dollars
- **Unsold Inventory Value** = brands with high end-inventory & low turnover (dead stock flag)

7. Project Roadmap

Phase	What	Status
0	Business problem & scope (this doc)	✅ Done
1	Load data into SQL Server	✅ Done (verified)
2	SQL: build Vendor Summary view (joins all 6 tables)	⏳ Next
3	Python (Jupyter/VS Code): pull summary, clean, engineer KPIs	Pending
4	Python: EDA + hypothesis testing (top vs. bottom vendors)	Pending
5	Power BI: dashboard + DAX measures	Pending
6	Insights & recommendations write-up	Pending